1. Do the extra data improve our predictions?

Professor, I have now separated two questions that were mixed together in the earlier presentation. First, what relationships and future outcomes does my Bayesian model consider plausible? Second, does adding a source of data actually improve predictions for later students? I implemented posterior sampling and a real MCMC check for the first question, and compared models on the same later students for the second. The central finding is that the extra data did not improve average error beyond the larger grade bundle. This is an exploratory result from existing classes, not proof that motivation or wellness is unimportant. The original notebook and original results remain available.

2. 191 students, six terms, four types of inputs

I have 191 distinct students from six course terms. I use assessment percentages, two logged LMS activity measures, three EVC-style survey features and wellness. The current table contains early expectancy, value and a cost field renamed cost_support. I have not revalidated its questionnaire scoring, so I do not interpret a positive cost score as automatically better or worse. All four early survey measures are missing for 78 students. The earlier notebook explicitly excluded a separate MLC RETRO export. That export and the raw ZIP are not present here, so I cannot claim that I tested retrospective responses. If those responses were collected after the forecast date, using them for an earlier prediction would leak later information.

3. A useful map does not establish predictive value

I still think the DAG is useful as a transparent map of assumptions, but a plausible drawing does not show that every data source improves forecasting. In the original full graph, all six non-grade predictors and all sixteen grade variables are direct parents of the final outcome. With complete inputs, that final conditional distribution is a Bayesian regression, so I can test it directly without changing the other arrows. The predictor graph matters when missing inputs are inferred. In this new controlled comparison I instead fill missing values with training medians in every model. This gives a consistent comparison, but it is not full Bayesian inference over missing inputs, and it is not a test of the original graph's causal correctness. Conditioning on grades may also remove pathways through which motivation matters.

4. Three different questions need three different checks

I interpret your suggestion about posterior probabilities in three ways. The first is a coefficient probability, such as the probability that the page-view association is positive after conditioning on grades and the other inputs. The second is a student outcome probability, such as the chance of finishing below an illustrative 70 percent. Both come from the Bayesian outcome model. The third question is whether adding a data group reduces prediction error. I test that using paired later-term comparisons. I also use a Bayesian bootstrap to describe uncertainty about reweighting the four observed term gains. That bootstrap probability is separate from the coefficient posterior and is not the probability that the model will improve in a new course term.

5. The sampler agrees with the exact Bayesian solution

I drew 20,000 independent samples from the exact conjugate posterior and also ran an actual two-block Gibbs sampler. Gibbs alternately updates the regression weights and residual variance. There are four chains, each with 1,000 warmup iterations followed by 4,000 saved iterations. The chains agree with the analytic posterior means and intervals. The largest rank-normalized R-hat is about 1.0007 and the smallest bulk effective sample size is above 12,000. Those checks support numerical accuracy for this model. They do not validate the assumed likelihood, graph, missing-data treatment or real-world forecasting performance. Since an exact solution exists, I would normally use it. MCMC becomes more useful if I add a robust likelihood, term effects or an explicit missing-data model.

6. Train on earlier terms; predict the next term

I evaluate Spring 2025, Fall 2025, Winter 2026 and Spring 2026 in order. Each time, training uses only terms before the evaluation term. Imputation values and scaling are fitted only on those training students. For each fold, I run the eight combinations of LMS, EVC and wellness in three contexts: no grades, the first grade bundle and the second grade bundle. That is 24 comparisons per fold, or 96 fits in the main analysis. I inherit coefficient precision 10 from the explored original workflow and check alternative priors rather than selecting a winner from these new results. The same records and settings have been examined before, and the training windows overlap, so I describe these as exploratory temporal checks, not independent confirmatory experiments. Grade-release dates remain unknown.

7. Adding all three sources did not meaningfully reduce error

The grade scores carry most of the predictive information in these comparisons. Without grades, the mean baseline has an average error of 6.54 points and all three extra sources give 6.52. That difference is only about 0.014 points. With the first grade bundle, error increases from 3.61 to 3.70. With the larger bundle, it increases from 2.95 to 3.04. The larger-bundle combined model is worse in all four term checks. This does not mean the extra sources are useless in every model or setting. It means this particular processed representation and model do not demonstrate meaningful incremental MAE benefit. These are new four-term macro results, not the original single Spring 2026 table of 2.82 versus 2.83.

8. LMS shows a modest signal when grades are absent

I do not want the full-model comparison to hide a useful source-specific result. Without any grades, LMS activity alone reduces macro average error from 6.54 to 6.36 points, a gain of 0.177 points. All four evaluation terms improve. This is a modest signal, and it also appears under the two alternative shrinkage strengths, although its size changes. LMS plus wellness gives a similar 0.198-point gain, so this does not establish a separate wellness contribution. EVC alone worsens MAE in all four checks. Adding every source together reduces the improvement to only 0.014 points. The no-grades setting is a controlled evidence scenario, not a verified early-course forecast. These are multiple exploratory comparisons on already examined records, so I would use the LMS finding to guide a dated follow-up analysis rather than declare it a selected winning model.

9. No single addition shows a stable Block B MAE gain

I also checked the sources individually and in pairs, rather than only comparing grades with the full model. Adding LMS alone, EVC alone or wellness alone gives a small negative average MAE gain in the larger grade context. Removing each group from the full model is also included in the output tables. The intervals shown here come from a Bayesian bootstrap over the four observed term mean gains. They describe sensitivity to how these four terms are weighted. They do not cover every possible future term or uncertainty in model development. In particular, when all four term gains are negative, every positive-weight bootstrap average is negative. A bootstrap probability of zero improvement then follows from the observed support; it is not proof that future improvement is impossible.

10. Small conditional associations after grades are included

After fitting the larger grade model to the earlier 148 students, the page-view coefficient has an 89 percent posterior probability of being positive. However, its mean association is only 0.34 final-percentage points per training standard deviation. Its probability of exceeding one point in absolute value is below one percent. The other five non-grade coefficients also have less than one percent probability of an absolute effect above one point under this particular model. These statements are conditional on the prior, median-filled design, other predictors and fixed linear model. Grades may mediate or already summarize earlier engagement and motivation. Correlation among predictors and measurement error also affect coefficients. These are not posterior edge-existence probabilities, and they do not show that wellness or motivation has no broader educational or causal importance.

11. A probability is useful—and can still be wrong

The model now reports an explicit probability distribution for each outcome. For one complete-input Spring 2026 student, the predicted mean is 71.24, the 90 percent interval is 65.72 to 76.75, and the probability of finishing below 70 is 35.6 percent. The actual final result was 75.95. For another student, whose surveys were missing and filled from training medians, the model assigned only a 2.3 percent probability below 70, but the actual outcome was 68.03. The 90 percent interval missed that result. A separate extreme outcome of 20.01 was predicted near 45.91, so the model also missed its severity. Integrating parameter uncertainty does not automatically fix those failures. These reused examples are diagnostics; they do not establish calibration or safe early-warning decisions.

12. The diagnosis survives several reasonable alternatives

I reran the comparisons with weaker and stronger coefficient shrinkage, with missingness flags, and on students with complete non-grade inputs. The mean gain from adding all three sources to the larger grade bundle remains negative in every version. It is about minus 0.19 with precision 1, minus 0.088 with precision 10, minus 0.007 with precision 100, minus 0.119 with missingness indicators, and minus 0.106 for respondents. The respondent analysis changes the population and does not solve selection bias. The no-grades result is more prior-sensitive: its full-model gain ranges from about minus 0.099 to plus 0.082 over precisions 1 to 100. I would not interpret those small changes as a robust practical benefit. The detailed output also includes log scores, interval coverage, widths and Brier scores.

13. Good exploratory framework; predictive claims need restraint

My assessment is that this is a good capstone framework for transparent exploratory prediction and uncertainty. It is not evidence that the specified graph is causal or that a coaching system will improve grades. Assessment marks are components of the final percentage, so their predictive strength is partly a part-whole relationship and may reflect late-course information. Missing survey responses, unknown release dates and only six terms limit the current analysis. The Gaussian model also misses some low outcomes and ignores uncertainty in imputed inputs. I would not add MCMC just to make the method sound more Bayesian. The sampler is already validated against a tractable reference. I would first verify measurement and timing, then consider a robust likelihood, term-level partial pooling or explicit missing-data modeling if those choices improve the diagnosed weaknesses.

14. Use the existing records to answer a sharper question

My next step is to recover the original data archive and verify when each grade and survey response became available. I would confirm the EVC and cost item wording and distinguish the early survey from the separate retrospective export. Then I would choose one useful forecast checkpoint and repeat the same paired source comparisons using only records available by that time. If retrospective responses occurred after that checkpoint, I would analyze them only as a separate retrospective association study. I would ask you to agree on a useful size of prediction gain and an appropriate risk threshold before interpreting probabilities as operational decisions. Finally, if low outcomes remain poorly represented, I would test a robust or hierarchical model against this fixed Bayesian baseline. An honest finding of no added predictive value is a valid research result.

15. One posterior; two sampling routes

The standardized response z equals A beta plus Gaussian noise. The coefficient prior is normal conditional on residual variance, with precision 0.001 for the intercept and 10 for slopes. The variance prior is inverse-gamma with shape 2 and scale 1. The posterior has precision L, mean m, shape a and scale b. Direct simulation draws variance from its inverse-gamma marginal and beta from its conditional normal. Gibbs draws beta conditional on the current variance, then variance conditional on beta. The second conditional adds p over two to the inverse-gamma shape and a quadratic coefficient deviation to the scale. For a new fixed design row, integrating both parameter types gives an exact Student-t predictive distribution. This is why exact simulation is available and MCMC is optional for this model.

16. Evaluate distributions as well as point errors

I score predictive distributions with negative log predictive density, where lower is better, and also report interval coverage, interval width and a Brier score for the illustrative below-70 event. I use the unbounded Student-t density and unbounded intervals; clipping is applied only to point predictions for MAE. Coverage near 90 percent on a small reused sample does not prove calibration, especially for low outcomes. The full model has slightly worse log score and slightly wider intervals, while its Brier score is only marginally lower. That mixed pattern does not justify claiming a clear distributional improvement. The original all-student DAG marginalized missing prediction inputs using fixed joint parameters. The new comparison fills those inputs and integrates outcome parameter uncertainty, so the two all-student procedures should not be described as identical.

17. What was run, and what remains unverified

The analysis used the existing processed table, whose hash and package versions are recorded in provenance.json. It fit all eight source combinations under three grade contexts across four later terms, repeated the experiment under four sensitivity settings, drew 20,000 exact posterior samples and ran four Gibbs chains. The executed notebook and tests document the run. The raw ZIP and separate MLC RETRO export remain unavailable, so extraction, survey reliability and retrospective-response timing are not newly verified. No posterior causal edge probabilities or deployed coaching claims are made. The methodological references identify standard Bayesian regression, Bayesian bootstrap and MCMC diagnostic methods; they are not sources of our empirical results.